```
arr=[10,25,30,45,50]
key=30
found=False
for i in range(len(arr)):
    if arr[i] == key:
        print("key found at index",i)
        found=True
        break
if not found:
        print("key not found")
```

```
key found at index 2

=== Code Execution Succe
```

main.py

Share Run

```python
def binary_search(arr, low, high, key):
    if low > high:
        return -1
    mid = (low + high) // 2
    if arr[mid] == key:
        return mid
    elif key < arr[mid]:
        return binary_search(arr, low, mid - 1, key)
    else:
        return binary_search(arr, mid + 1, high, key)
arr = [5, 10, 15, 20, 25]
key = 20
index = binary_search(arr, 0, len(arr) - 1, key)
if index == -1:
    print("Key not found")
else:
    print("Key found at index", index)
```

Output

```
Key found at index 3

=== Code Execution Successful ===
```

main.py

```python
n=5
fact=1
for i in range(1,n+1):
    fact*=i
print("Iterative:",fact)
def factorial(n):
    if n==0 or n==1:
        return 1
    return n*factorial(n-1)
print("Recursive:",factorial(n))
```

Output

```
Iterative: 120
Recursive: 120

=== Code Execution Successful ===
```

main.py

Share Run

```python
n=6
a,b=0,1
for i in range(n):
    print(a,end=" ")
    a,b=b,a+b
```

Output

```
0 1 1 2 3 5
=== Code Execution Successful ===
```

main.py

```python
A=[[1,2],[3,4]]
B=[[5,6],[7,8]]
result=[[0,0],[0,0]]
for i in range(2):
    for j in range(2):
        for k in range(2):
            result[i][j]+=A[i][k]*B[k][j]
print(result)
```

Output

```
[[19, 22], [43, 50]]

=== Code Execution Successful ===
```